

PS 26058 Compliance Matrix

CURRENT PROOF

ESTIMATED WATER PERFORMANCE

FINISHED-PRODUCT TARGET

This matrix does not omit projected capability. Each row states what exists now, what the completed payload can achieve, and the measurement that will close the requirement.

ID	PS clause	Where implemented	Verification method	Result: current and finished	Evidence	Status
C-01	"Design, prototype, and demonstrate a physical, self-contained Software-Defined Sonar Transmitter Payload Module."	AUV_Firmware/src/main.cpp; AUV_Hardware_Wiring_README_v2.md; Hyper Grey.stl	Inspection, enclosure geometry check, powered bench demonstration	Current: 1 ESP32-S3 bench stack; V1 enclosure 168 x 210 x 40 mm; 3,934-facet manifold mesh; 0 immersion tests. Finished: sealed AUV payload tray with wet interfaces and a 300 m design target.	C-01_physical-module.png	Partially met
C-02	"Use hardware timers and DMA to stream wave-generation arrays directly to a DAC without stalling the CPU."	SeaNergy/src/core/dsp.ts; SeaNergy/docs/07-physics-and-dsp.md; firmware architecture	Known-answer DSP checks, throughput calculation, future logic-analyser trace	Current: DDS/DMA architecture and app core verified; 0 hardware DMA waveform streams in main.cpp. Design rate: 1.178 MSa/s against a 1.25 MSa/s SPI ceiling. Estimated transmit-core idle: 75-95%; finished target: at least 80% measured.	C-02_dma-throughput.png	Partially met
C-03	"Support multiple modulation types on the fly: LFM chirps, geometric sweeps, and phase-coded pulses."	SeaNergy/src/core/dsp.ts; app/(tabs)/wave.tsx; app/(tabs)/spec.tsx	Mode-by-mode synthesis checks and render tests	Current: 4 software modes - CW, LFM, geometric, Barker-13 - pass finite-output checks; current hardware emits 1 fixed 1 kHz tone. Finished: all 4 modes switchable without recompilation and visible on scope/FFT.	C-03_waveform-modes.png	Partially met

Source basis: SIH PS 26058; SeaNergy code and tests; internal defence brief. Estimated figures require the stated hardware validation.

PS 26058 Compliance Matrix

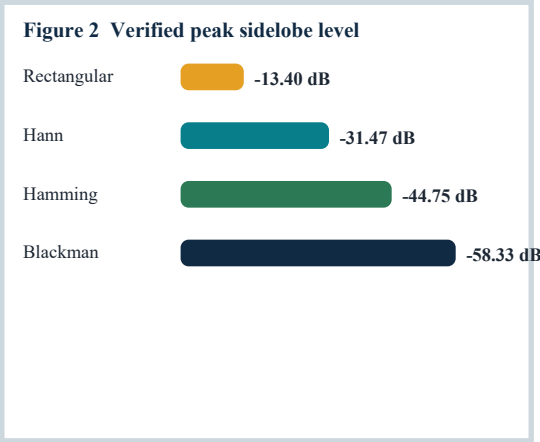
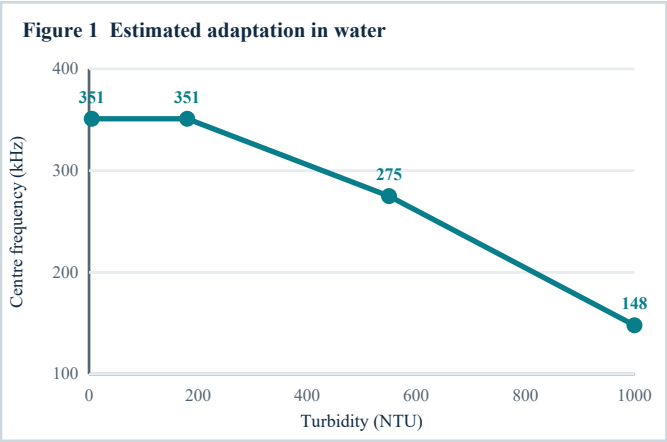
Status reflects the present build. Estimated and target figures remain included so judges can see the complete engineering intent without confusing projection with measurement.

ID	PS clause	Where implemented	Verification method	Result: current and finished	Evidence	Status
C-04	"Ingest real-time environmental data via physical sensors, or analog voltage dials acting as sensor inputs."	AUV_Firmware/src/main.cpp; SeaNergy/src/data/scenarios.ts; website/src/data/environment.ts	ADC/temperature log inspection and live potentiometer injection	Current: temperature, TDS, turbidity and 3 analog controls are read; 33 logged blocks exist. Finished: calibrated temperature, salinity, turbidity and depth channels with fault flags and recorded units.	C-04_sensor-interface.png	Partially met
C-05	"Modify bandwidth/center frequency, pulse duration, and amplitude/signal power instantly."	SeaNergy/src/core/adaptation.ts; website/src/data/measurements.ts	Turbidity sweep and solver-direction known-answer test	Modelled water sweep: 5 to 1,000 NTU moves centre frequency 351 to 148 kHz, bandwidth 299 to 96 kHz, and resolution 2.6 to 8.0 mm. Pulse duration reaches 7.4 ms. Amplitude candidates span 0.30-0.95; hardware amplitude control remains open.	C-05_adaptation-sweep.png	Partially met
C-06	"Use analog low-pass filters and an operational amplifier, combined with Hamming, Hann, or Blackman digital windows."	AUV_Firmware hardware chain; SeaNergy/src/core/dsp.ts; website/src/data/dsp.ts	FFT window measurement; future frequency-response, THD and SFDR sweep	Verified software PSL: rectangular -13.40 dB, Hann -31.47 dB, Hamming -44.75 dB, Blackman -58.33 dB. LPF/op-amp exists but is uncharacterised. Finished target: THD at most 1.0% and SFDR at least 55 dBc.	C-06_filter-window.png	Partially met

PS 26058 Compliance Matrix

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ID	PS clause	Where implemented	Verification method	Result: current and finished	Evidence	Status
C-07	"Validate clean, low-distortion, mathematically sound spectrograms on an oscilloscope or spectrum analyzer."	SeaNergy/src/core/___verify___.ts; SeaNergy/___tests___/screens.test.tsx; TX-001 to TX-004 plan	Scope capture, FFT, distortion measurement and automated regression	Current: 85 physics/core checks and 33 screen cases pass; 0 of 4 planned hardware scope tests are complete. Finished: verify 165.825-235.000 kHz LFM at 1.178 MSa/s, 23.81 dB compression gain, THD at most 1.0%, SFDR at least 55 dBc.	C-07_output-validation.png	Not met
C-08	"Focus heavily on low-power consumption and hardware-level optimization."	SeaNergy/src/core/power.ts; firmware DDS/DMA design; future INA226 trace	Model calculation followed by current-shunt integration over a ping cycle	Estimated coastal case: 5.58 mJ/ping and 7.26 h at a 10 s interval using a 2,600 mAh 3S pack with 15% derating. Transmitter duty is 0.5% in this case. Current-sense hardware is not yet fitted; finished target is within 10% of the model.	C-08_power-endurance.png	Partially met



Quick number	Current / target
Software verification	118 passed; 0 failures / retain regression gate
Power / endurance	5.58 mJ per ping; 7.26 h / water correlation within 10%
Cost	INR 2,500-4,500 / INR 36,000 marine-ready estimate

Source basis: SIH PS 26058; SeaNergy code and tests; internal defence brief. Estimated figures require the stated hardware validation.